

Converting LTL to Buchi

Wishnu Prasetya

wishnu@cs.uu.nl

www.cs.uu.nl/docs/vakken/pv

Converting LTL to Buchi

- Given an LTL formula φ , construct a Buchi automaton M that accepts the same sentences as φ .
 - Recall: “sentence” is a sequence of ‘symbols’, each is a set of propositions. Sentence = (abstract) execution.
- Steps:
 - Construct GNBA
 - Convert to NBA
 - Optimize

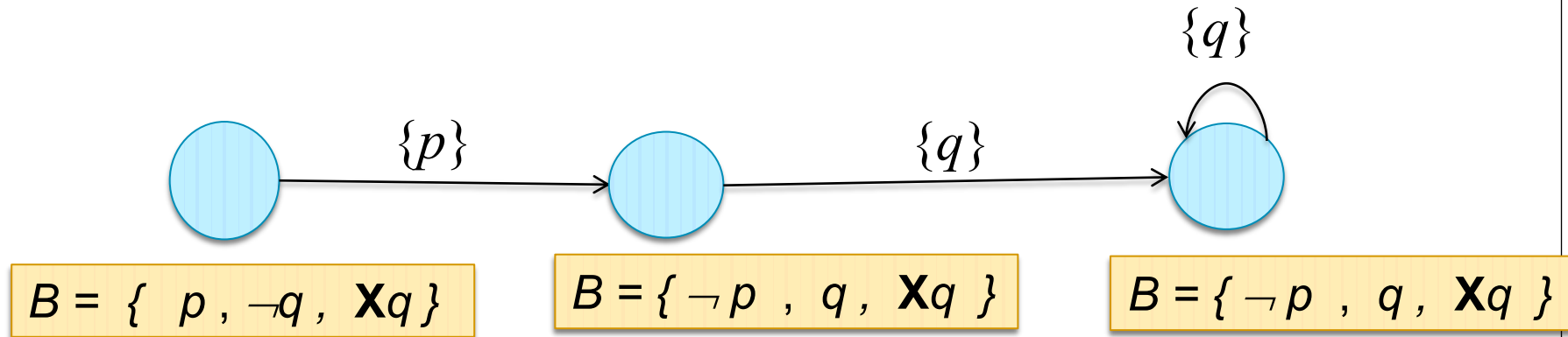
Restricting to X/U

- All LTL formulas can be expressed with just **X** and **U**.

$$\begin{aligned}\langle \rangle \varphi &= \text{true } \mathbf{U} \varphi \\ \Box \varphi &= \neg (\langle \rangle \neg \varphi) \\ \varphi \mathbf{W} \psi &= \Box \varphi \vee \varphi \mathbf{U} \psi\end{aligned}$$

- Let's assume that your input formula is expressed in this form of LTL.

Idea..



To help us, each state s will be labeled with an “observation” B . It is a consistent set of formulas. Any infinite sequence starting from s must satisfy all formulas in B .

The set of candidate “observations” for a given ϕ is finite; and we can figure out how to connect them with arrows.

Closure

- **closure**(φ) is the set of all
 - subformulas of φ (incl φ itself)
 - negations of subformulas
- Example: $\varphi = p \mathbf{U} q$

$$\mathbf{closure}(\varphi) = \{ p, q, \neg p, \neg q, p \mathbf{U} q, \neg(p \mathbf{U} q) \}$$

- Only the value of the formulas in the closure can affect the value of φ .

Observation

- Example: $\varphi = p \mathbf{U} q$

$$\mathbf{closure}(\varphi) = \{ p, q, \neg p, \neg q, p \mathbf{U} q, \neg(p \mathbf{U} q) \}$$

- An '*observation*' B is in principle a subset of the closure, but we want it to be 'consistent' and 'maximal'.
 - $\{ p, q, p \mathbf{U} q \} \rightarrow \text{OK}$
 - $\{ p, \neg p \} \rightarrow \text{inconsistent}$
 - $\{ p \} \rightarrow \text{not maximal}$

Consistency of the B 's

- An observation B must be consistent with respect to propositional logic:
 - f and $\neg f$ cannot be both in B
 - $f \wedge g \in B \Rightarrow f, g \in B$
- Consistent with respect to “until”. For any $f \mathbf{U} g \in \mathbf{closure}(\varphi)$:
 - $g \in B \Rightarrow f \mathbf{U} g \in B$
 - $f \mathbf{U} g \in B$ and $g \notin B \Rightarrow f \in B$

Maximality

- Every observation B should be *maximal* $\rightarrow$

For every $f \in \mathbf{closure}(\varphi)$, either $f \in B$ or $\neg f \in B$.

- Ex.

$$\varphi = p \mathbf{U} q$$

$$\mathbf{closure}(\varphi) = \{ p, q, \neg p, \neg q, p \mathbf{U} q, \neg(p \mathbf{U} q) \}$$

5 Observations
(blue). Red ones
may look like
observations, but
are inconsistent.

$\{ p, q, p \mathbf{U} q \}$

$\{ p, q, \neg(p \mathbf{U} q) \}$

$\{ p, \neg q, p \mathbf{U} q \}$

$\{ p, \neg q, \neg(p \mathbf{U} q) \}$

$\{ \neg p, q, p \mathbf{U} q \}$

$\{ \neg p, q, \neg(p \mathbf{U} q) \}$

$\{ \neg p, \neg q, p \mathbf{U} q \}$

$\{ \neg p, \neg q, \neg(p \mathbf{U} q) \}$

Constructing the automaton A_φ

- **States:** observations from **closures**(φ)
- **Initial states:** all states that contain φ
- **Arrows:** for any pairs observations B, C add this arrow:

$$B \text{ --- } V \longrightarrow C$$

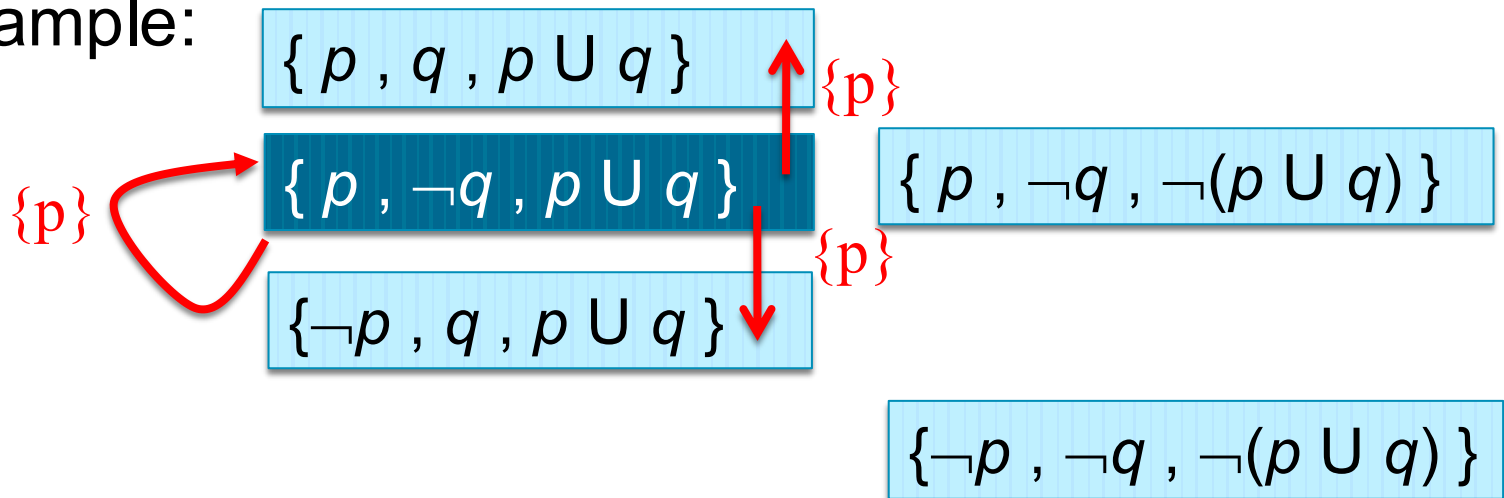
- V = the set of positive (not in negation) propositions in B .
 - If this arrow is 'consistent'
- **Acceptance** states?

The arrows

- $B \text{ --- } V \longrightarrow C$ is consistent if (1) :

- $\mathbf{X}f \in B \Rightarrow f \in C$
- $f \mathbf{U} g \in B \Rightarrow g \in B$
or $(f \in B \text{ and } f \mathbf{U} g \in C)$

- Example:



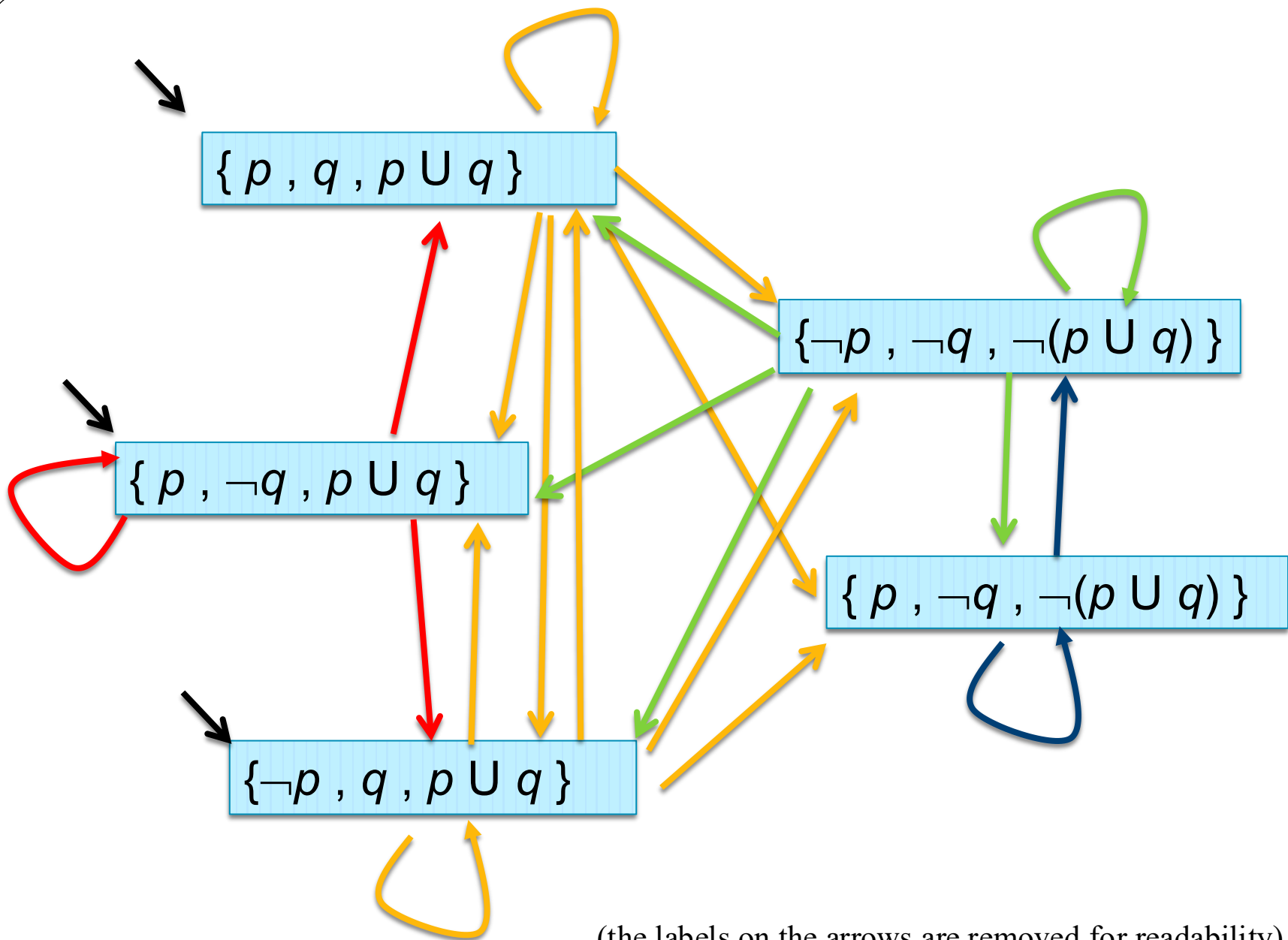
The arrows

- $B \multimap V \longrightarrow C$ is consistent if (1) :

- $\mathbf{X}f \in B \Rightarrow f \in C$
- $f \mathbf{U} g \in B \Rightarrow g \in B$
or $(f \in B \text{ and } f \mathbf{U} g \in C)$

- Furthermore (2) :

- $\neg \mathbf{X}f \in B \Rightarrow \neg f \in C$
- $\neg(f \mathbf{U} g) \in B \Rightarrow$
 $(\neg f \in B \text{ and } \neg g \in B)$
or
 $(f \in B \text{ and } \neg g \in B \text{ and } \neg(f \mathbf{U} g) \in C)$



(the labels on the arrows are removed for readability)

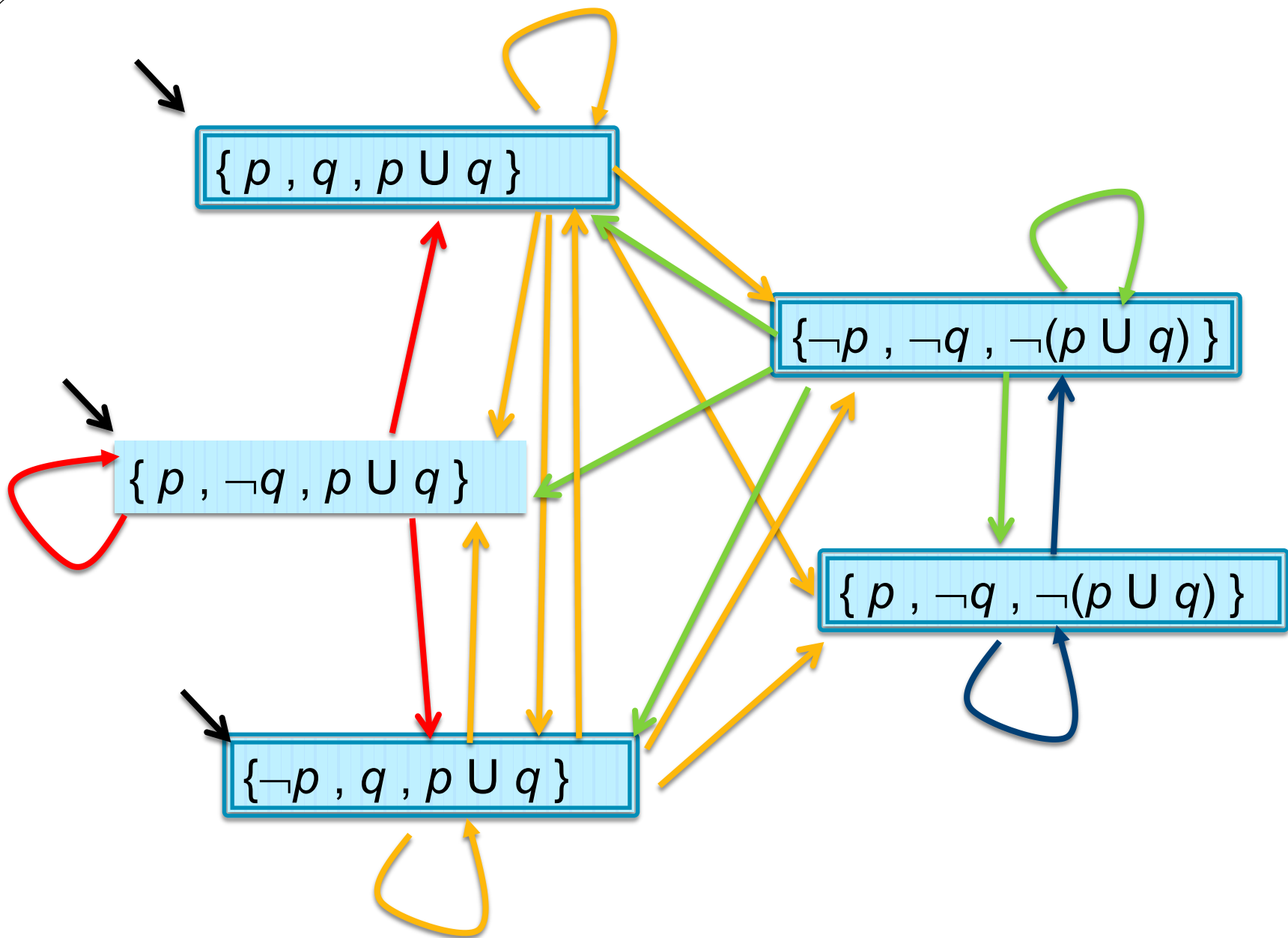
Enforcing eventuality

- For each $f \mathbf{U} g \in \mathbf{closure}(\varphi)$, add an accepting group:

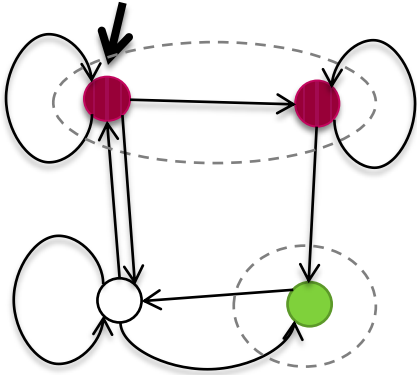
$$\mathbf{F} (f \mathbf{U} g) = \{ B \mid B \in \mathbf{Q} \wedge g \in B \} \cup \{ B \mid B \in \mathbf{Q} \wedge f \mathbf{U} g \notin B \}$$

where $\mathbf{Q}$ is the set of states of GNBA of φ that we are constructing.

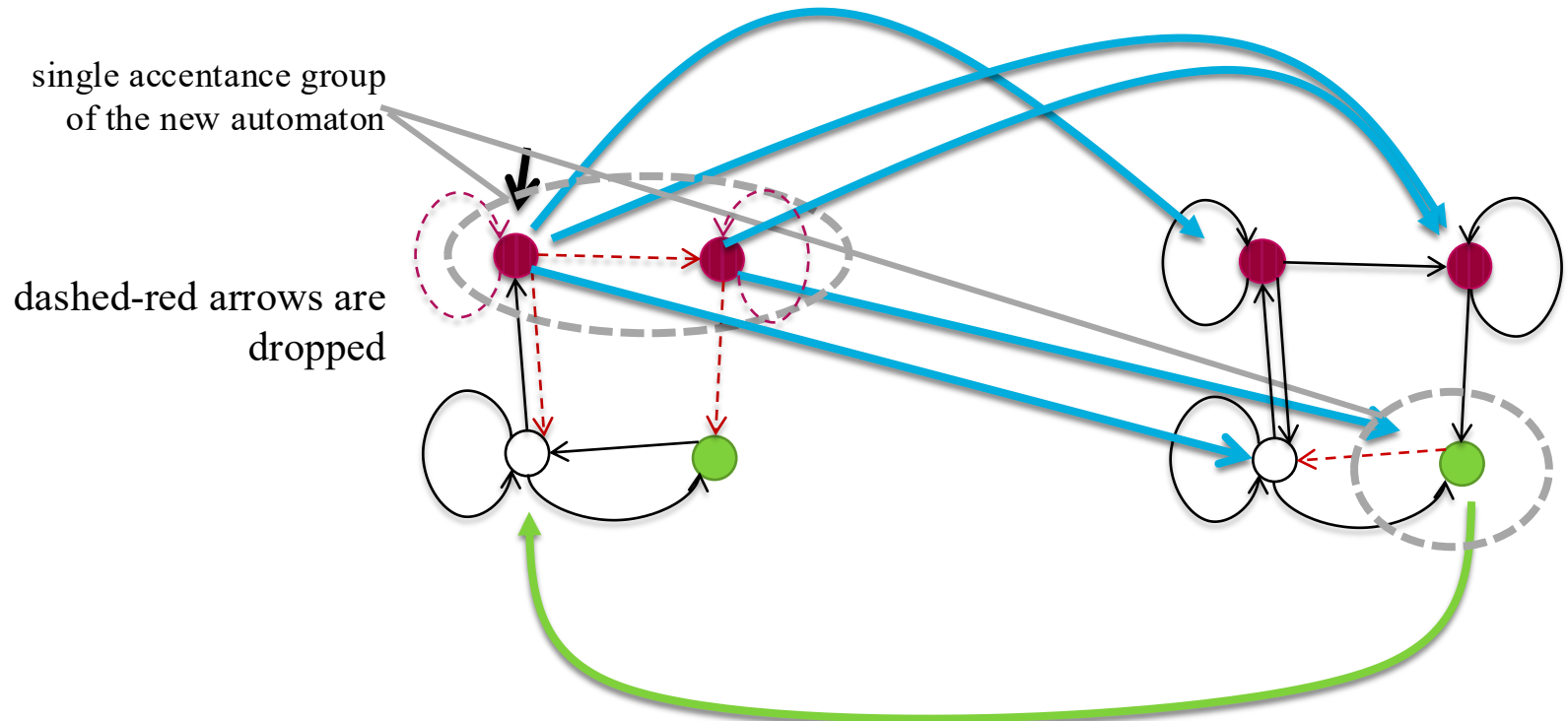
($\mathbf{Q}$ = the set of all ‘observations’)



From GNBA to NBA



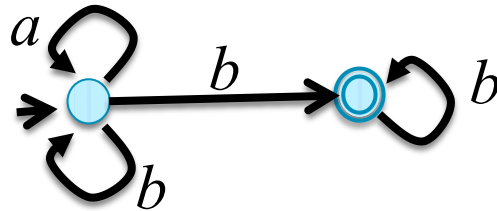
GNBA with 2x accepting groups.



Can we make it deterministic?

- In ordinary automaton, DFA can be converted to an equivalent NDFA (equivalent = generating the same sentences).

- For Buchi?



No *deterministic* Buchi can generate the sentences of this Buchi

- NBA is really more powerful than DBA.

How big are they?

- NDGBA generated by our procedure $\rightarrow |M| = 2^{|\varphi|}$.
- Converting to NDBA multiplies the number of states with C , where C is the number of **U** in φ